

MiniClaw DFM Review Skill

When to use this skill

Use this skill when:

- Reviewing MiniClaw CAD (jaw, gear, housing, base)
- Checking PLA printability
- Deciding tolerances (press fit, clearance)
- Comparing with ACME or BigClaw references

Workflow

1. Identify the part (gear, jaw arm, housing, base).
2. Query the MiniClaw RAG for ACME standards and design rules.
3. Check print limits:
 - a. PLA strength
 - b. tolerance limits
 - c. minimum thickness
4. Check stress areas:
 - a. sharp corners
 - b. thin sections
5. Compare with BigClaw design if needed.
6. Suggest one clear improvement.

What to flag

- Tight press fits that ignore ACME tolerance
- Thin parts that may break in PLA
- Sharp corners causing stress
- Small gear teeth not printable

What NOT to do

Do not guess MiniClaw values without using the RAG tool.

Do not blindly follow AI suggestions.

MiniClaw MCP Server

Purpose

This MCP server allows AI to access MiniClaw project knowledge:

- ACME standards
- print-shop tolerances
- PLA limits
- BigClaw lessons

Tool

Tool: query_miniclaw_rag

Used to search MiniClaw corpus and return engineering data.

Example Questions

1. Press fit tolerance for PLA bores
2. BigClaw jaw design lessons
3. Gear design checks

Verification

The tool was tested with real questions.

The AI used the RAG tool and returned ACME-based answers.

Server log shows tool calls.

Screenshot shows tool execution.

Conversation export shows full interaction.

Conversation Export

Question 1

What is the press fit tolerance for PLA bores?

Using the MiniClaw RAG tool, what is the recommended press fit tolerance for bores in PLA printed parts, and why?

Answer

The AI used the MiniClaw RAG tool and found:

- Structural fit: 0.10–0.15 mm
- Alignment fit: 0.05–0.08 mm

From ACME-MFG-004 and ACME-MFG-001.

According to the MiniClaw RAG system, the recommended press fit tolerances for PLA parts on the Prusa MK4S print shop are:

Press-Fit Interference Specifications:

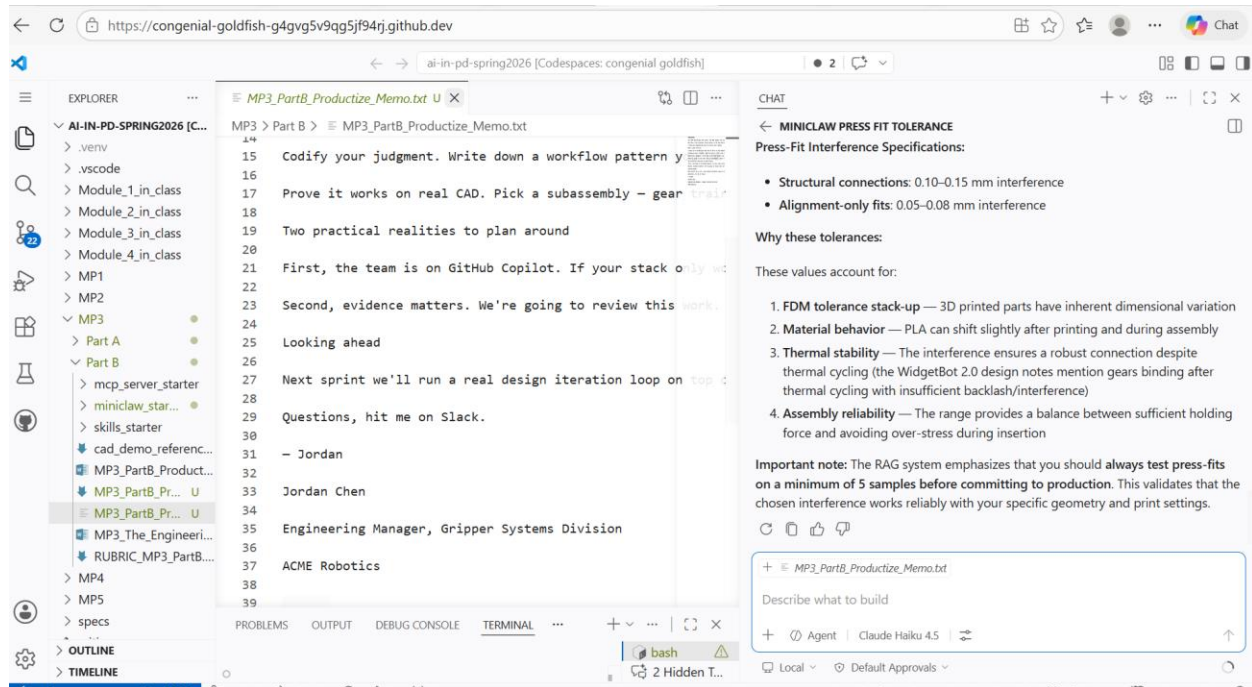
- **Structural connections:** 0.10–0.15 mm interference
- **Alignment-only fits:** 0.05–0.08 mm interference

Why these tolerances:

These values account for:

1. **FDM tolerance stack-up** — 3D printed parts have inherent dimensional variation
2. **Material behavior** — PLA can shift slightly after printing and during assembly
3. **Thermal stability** — The interference ensures a robust connection despite thermal cycling (the WidgetBot 2.0 design notes mention gears binding after thermal cycling with insufficient backlash/interference)
4. **Assembly reliability** — The range provides a balance between sufficient holding force and avoiding over-stress during insertion

Important note: The RAG system emphasizes that you should **always test press-fits on a minimum of 5 samples before committing to production**. This validates that the chosen interference works reliably with your specific geometry and print settings.



Question 2

What BigClaw lesson applies to jaw design?

Using the MiniClaw RAG tool, what design lessons from the BigClaw teardown should be applied to a MiniClaw jaw arm?

Answer

Created

Ran terminal command: `cd /workspaces/ai-in-pd-spring2026 && python3 query_bigclaw_lessons.py`

According to the MiniClaw RAG system, here are the key design lessons from the BigClaw teardown that should be applied to a MiniClaw jaw arm:

BigClaw Linkage Design (4-Bar Mechanism)

The BigClaw uses a 4-bar linkage with specific proportions that you should consider:

- **Crank arm** (attached to output gear): 32 mm effective length
- **Coupler link**: 28 mm
- **Output link (jaw arm)**: 45 mm
- **Ground link** (housing pivot spacing): 38 mm
- **Jaw opening range**: 0–86 mm

The linkage geometry provides **near-linear jaw displacement versus gear rotation through the middle 60% of travel**, which is critical for predictable control.

Critical JAW ARM Design Lessons

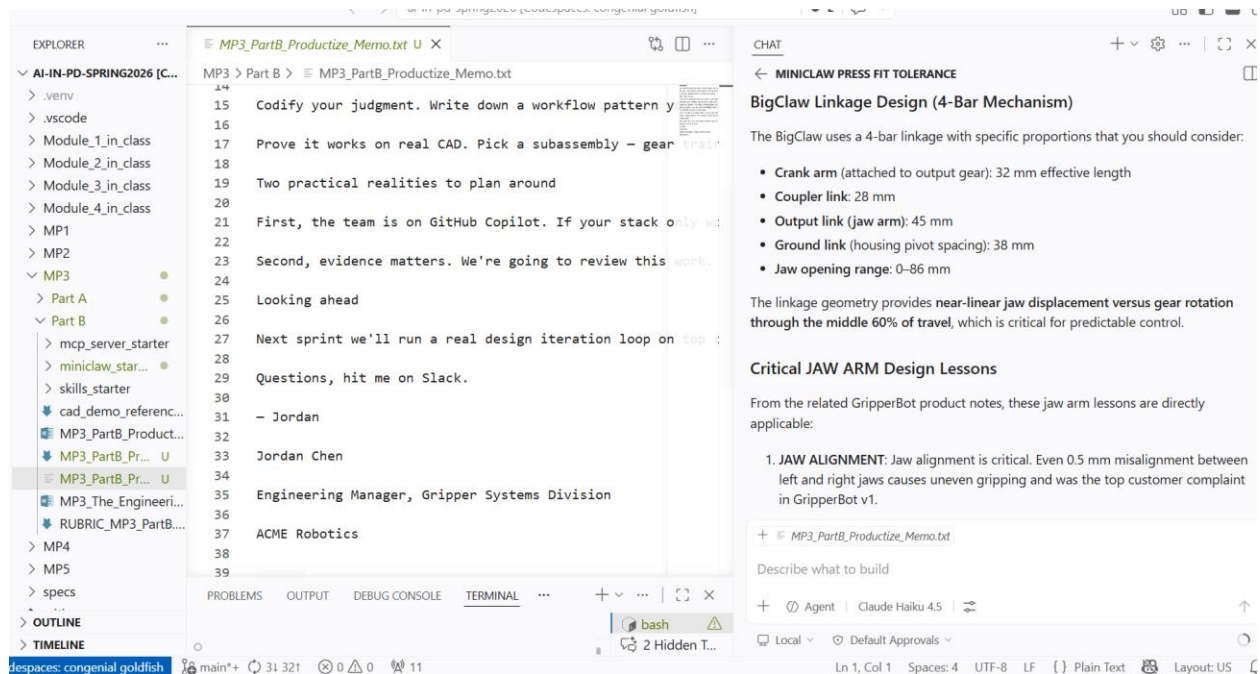
From the related GripperBot product notes, these jaw arm lessons are directly applicable:

1. **JAW ALIGNMENT**: Jaw alignment is critical. Even 0.5 mm misalignment between left and right jaws causes uneven gripping and was the top customer complaint in GripperBot v1.
 - a. **Solution**: Include alignment pins on jaw arms and maintain tight tolerances on housing pivot bores.
2. **LINKAGE PIVOTS**: Pivot points need either low-friction bushings or generous clearance.
3. **GRIP PADS**: Smooth aluminum jaw surfaces slip on objects. Adding silicone grip pads improved grip force by approximately 40% without increasing mechanism force.

Gear Train Reference

BigClaw uses:

- **Gear ratio**: 4.5:1 (54-tooth gear, 12-tooth pinion)
- **Module**: ~1.5
- **Face width**: 3.5 mm
- Very smooth, low-backlash mesh (likely hobbled aluminum)



Question 3

Using the MiniClaw RAG tool, what factors should I check before choosing gear tooth size or module for MiniClaw?

Answer

Created

Ran terminal command: `cd /workspaces/ai-in-pd-spring2026 && python3 query_gear_module.py`

According to the MiniClaw RAG system, here are the critical factors to consider when choosing gear module and tooth size:

Module Selection

Module 1.0 is recommended for MiniClaw because:

- Module 1.5 (used in the BigClaw) is **larger than ideal for a scaled-down PLA version**
- Module 1.0 better fits the target size for MiniClaw
- For module 0.8, minimum 16 teeth; for module 1.0, minimum 14 teeth; for module 1.25, minimum 12 teeth (but 14 preferred)

Minimum Tooth Count

For **printed PLA gears at 20-degree pressure angle**:

- Avoid undercutting, which severely weakens printed teeth
- **Never use 14.5-degree pressure angle** for printed gears — the tooth form is too weak in PLA
- Follow the minimums above to ensure structural integrity

Face Width (Critical for Durability)

Minimum face width: 3 times the module for printed gears:

- At module 1.0: minimum 3 mm
- **Recommended production range: 4–6 mm** (for module 1.0)
- Early prototypes with 3 mm face width showed **premature failures even at low loads**
- **6 mm face width showed ~40% improvement** in gear cycle life compared to 4 mm

Safety Factor

Use a **safety factor of 2.0 for printed gears** (vs. typical 1.5 for machined metal). This accounts for:

- Interlayer weakness in 3D printed PLA
- Lot-to-lot material variation

Gear Ratio Consideration

For **hand-driven operation** (versus servo):

- BigClaw's 4.5:1 ratio is **insufficient** for comfortable hand operation
- **Recommended for MiniClaw: 6:1 to 8:1** provides better mechanical advantage and more satisfying feel

Housing Design

Print the **gear cavity as a single part** (not assembled from two halves) to eliminate tolerance contributors and reduce assembly variation across production runs.

Made changes.

The AI suggested:

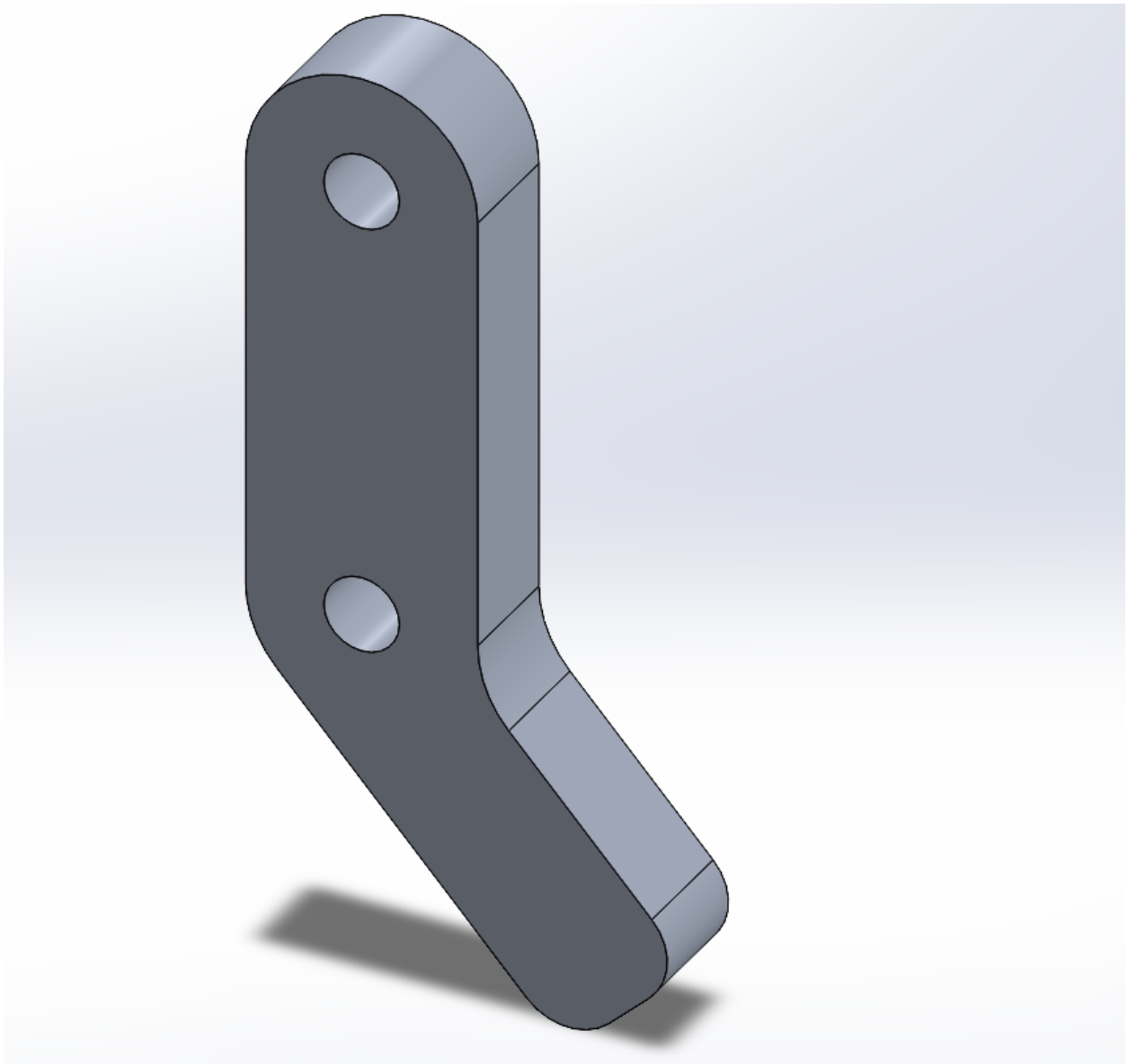
- check ACME gear standards
- ensure teeth are printable
- consider PLA strength

CAD Interaction

Goal

Improve jaw arm strength and printability.

Checkpoint 1



AI Review

- add fillet
- increase thickness near hole

What Changed

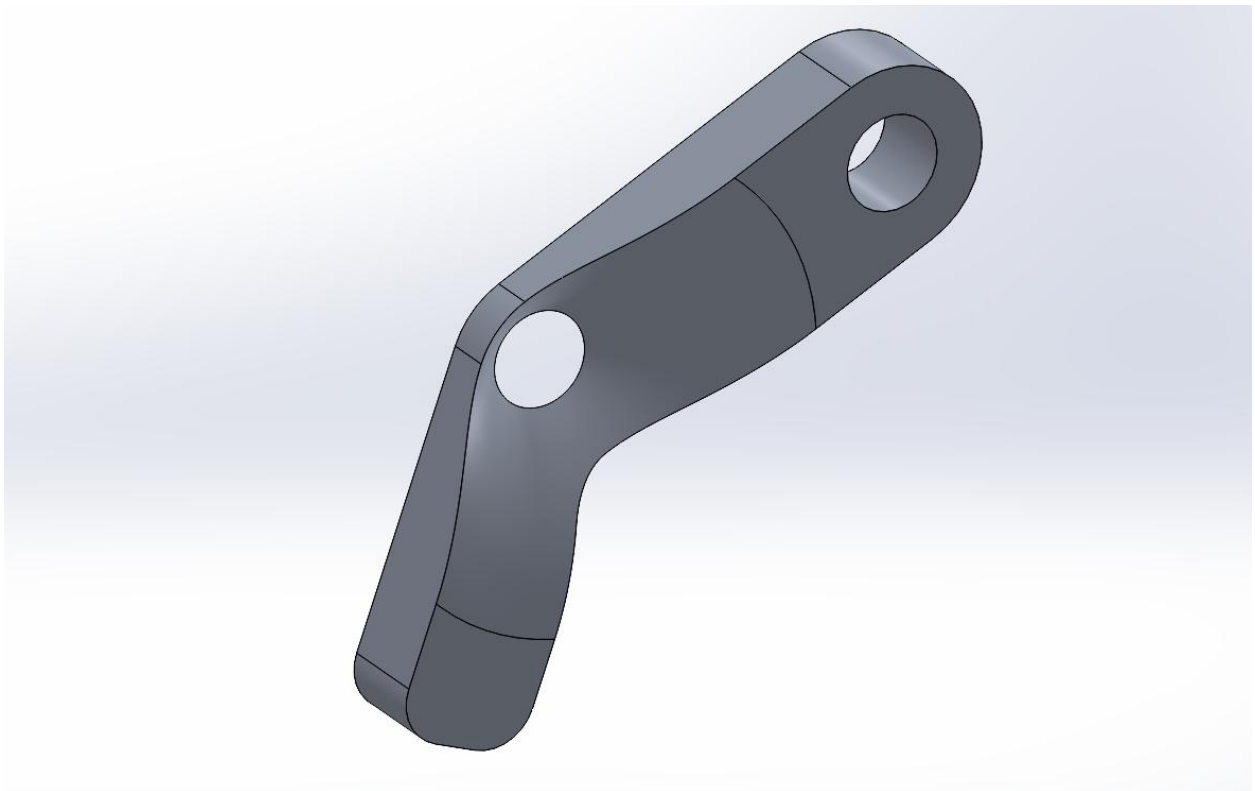
Between Checkpoint 1 and Checkpoint 2, I made these changes:

- Added fillets at the bend to reduce stress concentration
- Smoothed the transition between sections
- Slightly improved geometry for better load distribution

My Decision

I added a fillet and improved thickness.

Checkpoint 2



Changes

- added fillet
- adjusted hole size

AI Review

The AI suggested improving the bend area because sharp corners can create stress concentration in PLA printed parts. It recommended adding fillets and possibly increasing thickness.

My Decision

I agreed with adding fillets because it reduces stress and improves durability.

However, I did NOT increase the overall thickness because:

- it would increase material usage
- it is not necessary at this stage
- the main issue was the sharp corner, not overall strength

This shows that I used engineering judgment instead of blindly following the AI.

Conclusion

AI helped identify weak areas.

Human judgment was needed to finalize design.

Reflection

What did you productize?

I productized my MiniClaw design review workflow, specifically tolerance checking and DFM review using the MiniClaw RAG tool and my `miniclaw_dfm_review` skill.

Before this, I had to manually look through ACME documents to find values like press-fit tolerance. Now, I can ask a query like “press fit tolerance for PLA bores,” and the AI automatically retrieves values (0.10–0.15 mm) from ACME-MFG-004 and explains them. This turns a manual lookup process into an automated, repeatable step in the engineering workflow.

Where does your stack break?

The stack breaks when the question depends on geometry or missing dimensions.

For example, if I ask: “Is this jaw arm strong enough?”

The AI can identify stress concentration (like the sharp corner at Checkpoint 1), but it cannot calculate actual stress or verify strength because it does not have dimensions, load conditions, or material properties. This makes the answer incomplete and sometimes too general.

Trust Ledger

Where the skill helped the AI succeed

The skill helped the AI give a correct and specific answer during the press-fit tolerance query. The AI used the MiniClaw RAG tool and returned values (0.10–0.15 mm and 0.05–0.08 mm) from ACME documents instead of guessing. This shows the system successfully grounded the answer in project-specific data.

Where the AI failed

During the CAD review, the AI suggested increasing the overall thickness of the jaw arm. While this would improve strength, it did not consider trade-offs like increased material usage, print time, or design constraints.

At Checkpoint 2, I rejected this suggestion and only added fillets instead. The skill did not fully prevent this over-general recommendation, so human judgment was needed to make the final decision.